

RaceDay API Endpoint Plan

Purpose

This planning document defines the REST endpoints that the RaceDay system is expected to expose in Part 2. It is designed before API implementation and aligns with the database model in `RaceDay_ERD.png` and `RaceDay_Database.sql`.

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
POST	<code>/api/auth/register</code>	Registers a new participant or organiser account after validating the email and role.	Public	<code>firstName</code> , <code>lastName</code> , <code>email</code> , <code>password</code> , <code>role</code>	201 Created with <code>userId/token</code> ; 400 validation; 409 duplicate email.
POST	<code>/api/auth/login</code>	Authenticates a user and returns an access token.	Public	<code>email</code> , <code>password</code>	200 OK <code>token/user</code> ; 401 invalid credentials.
GET	<code>/api/profile</code>	Returns the logged-in user's profile.	Logged-in	None	200 OK profile; 401 unauthorised.
PUT	<code>/api/profile</code>	Updates the logged-in user's profile and contact details.	Logged-in	Profile fields	200 OK updated profile; 400 validation.
GET	<code>/api/events</code>	Lists available events with optional filters.	Public	Query parameters: <code>date</code> , <code>status</code>	200 OK event collection.

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
GET	/api/events/{id}	Returns one event and its categories.	Public	None	200 OK event; 404 not found.
POST	/api/events	Creates a race event for the logged-in organiser.	Organiser	name, description, venue, eventDate, registrationOpen, registrationClose, status	201 Created; 400 validation; 403 Open, Close, Close, Close.
PUT	/api/events/{id}	Updates an event owned by the organiser.	Organiser	Editable event fields	200 OK; 403 forbidden; 404 not found.
DELETE	/api/events/{id}	Deletes an event that has not started and is owned by the organiser.	Organiser	None	204 No Content; 403 forbidden; 404 not found; 409 conflict.
GET	/api/categories	Lists all race categories.	Public	None	200 OK category collection.
POST	/api/events/{eventId}/categories	Associates a category to an event.	Organiser	categoryId, entryFee	201 Created; 404 not found; 409 duplicate.
DELETE	/api/events/{eventId}/categories/{categoryId}	Removes a category when it has no enrolments.	Organiser	None	204 No Content; 409 conflict.

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
POST	/api/events/{eventId}/enrolments /api/enrolments	logged-in participant in an event category.	Participant	categoryId	201 Created with race number; 400 invalid; 409 closed/full/duplicate.
GET	/api/enrolments	Returns the logged-in participant's enrolments.	Participant	None	200 OK enrolment collection.
GET	/api/events/{eventId}/enrolments	enrolments for an organiser's event.	Organiser	None	200 OK; 403 forbidden; 404 not found.
DELETE	/api/enrolments/{id}	participant enrolment before the deadline.	Participant	None	204 No Content; 404 not found; 409 deadline passed.
GET	/api/events/{eventId}/results	published event results ordered by position.	Participant	None	200 OK result collection.
POST	/api/events/{eventId}/results	finishing result for an enrolment.	Organiser	enrolmentId, finishTime, position, resultStatus	201 Created; 400 invalid; 403 forbidden; 409 duplicate.

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
PUT	/api/results/ id	Creates an existing result.	Organiser	finishTime, position, resultStatus	200 OK; 403 forbidden; 404 not found.
GET	/api/results/ id	Fetches one published result.	Public	None	200 OK result; 404 not found.

Role model

- **Organiser:** creates and manages events, configures categories, views event enrolments and records results.
- **Participant:** maintains a profile, views events/categories, enrolls in events and cancels their own enrolments before the deadline.

Consistency rule

Part 2 implementation should closely follow these routes. Any deliberate deviation should be documented in the project README.